Essential in Revit structure (40 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	Introduction to Autodesk Revit Structure About Structural Elements Structural Templates Starting a new project GUI Placing structural objects as per guidelines Drawing aids Project units Linking CAD files. Linking and importing Revit files	Theory/Demo	2.0 Hrs
2	Session 2	Excercises included for practice	Practical	2.0 Hrs
3	Session 3	**Levels:** Structural objects - adding up levels, Creating levels, Modifying levels. **Template Creation:** Creating view templates, Applying and assigning view templates, Understanding visibilityand graphics overrides. **Grid :** Creating grids, Modifying grids	Theory/Demo	2.0 Hrs
4	Session 4	Excercises included for practice	Practical	2.0 Hrs
5	Session 5	**Structural Elements:** Modeling structural columns, Modifying structural columns, Placing structural columns asper drawing, Placing structural beams, Modifying structural beams, Beam – column joins **Structural Framing:** Modeling structural beam system, Modifying structural beam system **Structural walls & floors:** Creating structural walls, Modifying structural walls, Creating structural floors,Modifying structural floors	Theory/Demo	2.0 Hrs
6	Session 6	Excercises included for practice	Practical	2.0 Hrs
7	Session 7	**Steel Modeling:** Placing steel columns and beams, Placing bracing system, Modifying bracing system, placingstructural trusses, Modifying structural trusses **Structural Foundations:** About structural foundation walls, Placing and modifying structural foundation walls,About wall footings, Modeling types of footings, Modifying wall footings	Theory/Demo	2.0 Hrs
8	Session 8	Excercises included for practice	Practical	2.0 Hrs
9	Session 9	**Structural slabs & openings:** About structural slabs, modeling structural slabs as per load distribution, modelingstructural slabs, creating openings on structural elements. **Structural reinforcement:** About structural reinforcement, Adding rebars on structural elements, Modifyingrebars, Reinforcing walls, floors and slabs	Theory/Demo	2.0 Hrs
10	Session 10	Excercises included for practice	Practical	2.0 Hrs
11	Session 11	**Structural Analysis – Analytical model:** Placing loads, Boundary conditions and settings, Preparing projects forstructural analysis, Viewing analytical projects, Adjusting analytical projects	Theory/Demo	2.0 Hrs
12	Session 12	Excercises included for practice	Practical	2.0 Hrs
13	Session 13	**Creating views:** Duplicating views, Adding callout views, Elevations and sections **Creating Details:** Setting up detail views, Adding detail components, Annotating details **Construction Documents:** Working with dimensions, Working with text adding tags, Adding detail lines andsymbols, Creating legends, Setting up sheets, Placing and modifying view on sheets	Theory/Demo	2.0 Hrs
14	Session 14	Excercises included for practice	Practical	2.0 Hrs
15	Session 15	**Scheduling:** Structural schedules, Graphic column schedules, Working with schedules, Importing and exportingschedules **Family Creation:** Column creation, Beam creation	Theory/Demo	2.0 Hrs
16	Session 16	Excercises included for practice	Practical	2.0 Hrs
17	Session 17	**Family creation:** Creation of piles, Creating truss **Exporting Files:** Exporting to CAD formats, Exporting to IFC, Exporting to DWF/Dxf, Printing sheets	Theory/Demo	2.0 Hrs
18	Session 18	Excercises included for practice	Practical	2.0 Hrs
19	Session 19	**Project:** Structure final project	Theory/Demo	2.0 Hrs
20	Session 20	Excercises included for practice	Practical	2.0 Hrs